

FLUID MECHANICS (උවස්ථිතිය)

Advanced Level Science for Technology (SFT) - TechHub LK Study Guide

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1. Introduction to Fluids

Among the three states of matter (solid, liquid, gas), only **liquids** and **gases** are classified as **fluids** [cite: 7].

Fluids are broadly categorized into two types [cite: 7]:

- **Compressible Fluids:** Fluids whose volume can be decreased by applying external pressure (Gases) [cite: 7].
- **Incompressible Fluids:** Fluids whose volume cannot be decreased by applying external pressure (Liquids) [cite: 7].

This unit studies the behavior of these fluids and their related calculations [cite: 7].

2. Density (*ho*)

Density is defined as mass per unit volume [cite: 7].

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}} \implies \rho = \frac{m}{V}$$

- **SI Units:** $\frac{\text{kg}}{\text{m}^3}$ or $\frac{\text{kg}}{\text{m}^3}$ [cite: 7].
- Density is also commonly expressed in $\frac{\text{g}}{\text{cm}^3}$ [cite: 7].

Relative Density

Relative density is the ratio of the density of a substance to the density of water [cite: 7]. (Note: Density of water = $1000 \frac{\text{kg}}{\text{m}^3}$) [cite: 7].

$$\text{Relative Density} = \frac{\text{Density of Substance}}{\text{Density of Water}}$$

Worked Example 1

A metal cube with a side length of **10 cm** has a mass of **2 kg** [cite: 7].

1. Find the density of the metal.
2. Find the relative density of the metal.
3. What is the volume of **0.5 kg** of this metal?
4. What is the mass of **100 cm³** of this metal?

Solutions:

1. Volume of cube = $\text{side}^3 = 10\text{ cm} \times 10\text{ cm} \times 10\text{ cm} = 1000\text{ cm}^3 = 1 \times 10^{-3}\text{ m}^3$.

$$\rho = \frac{m}{V} = \frac{2\text{ kg}}{1 \times 10^{-3}\text{ m}^3} = 2000\text{ kg/m}^3 \text{ [cite: 7].}$$

2. Relative Density = $\frac{2000\text{ kg/m}^3}{1000\text{ kg/m}^3} = 2$
(dimensionless) [cite: 7].

3. Volume = $\frac{\text{Mass}}{\text{Density}} = \frac{0.5\text{ kg}}{2000\text{ kg/m}^3} = 2.5 \times 10^{-4}\text{ m}^3$ [cite: 7].

4. Mass = $\text{Density} \times \text{Volume} = 2000\text{ kg/m}^3 \times 100 \times 10^{-6}\text{ m}^3 = 0.2\text{ kg}$ [cite: 7].

3. Pressure

Pressure is defined as the force acting normally per unit cross-sectional area [cite: 7].

$$P = \frac{\text{Normal Force } (F)}{\text{Cross-sectional Area } (A)}$$

- **SI Units:** N/m^2 or N/m^2 , also expressed as Pascals (Pa) [cite: 7].
($1\text{ Pa} = 1\text{ N/m}^2$) [cite: 7].

- Always use perpendicular forces acting on the surface [cite: 7].
- If an oblique force is applied, resolve it to obtain its perpendicular component [cite: 7].

Worked Examples on Pressure

- **Example 2:** Force = 10 N , Area = $0.1\text{ m} \times 0.1\text{ m} = 0.01\text{ m}^2$.

$$P = \frac{10}{0.01} = 1000\text{ Pa} \text{ [cite: 7].}$$

- **Example 3:** Net downward force = $(30 - 5)\text{ N} = 25\text{ N}$, Area = $0.05\text{ m} \times 0.05\text{ m} = 0.0025\text{ m}^2$.

$$P = \frac{25}{0.0025} = 10,000\text{ Pa} \text{ [cite: 7].}$$

- **Example 4 (Inclined Force):** Force = $10\text{ N} \cos 60^\circ = 5\text{ N}$, Area = $0.2\text{ m} \times 0.2\text{ m} = 0.04\text{ m}^2$.

$$P = \frac{5}{0.04} = 125\text{ Pa} \text{ [cite: 7].}$$

4. Pressure at a Point in a Fluid

The pressure at a depth inside a stationary fluid is given by [cite: 7]:

$$P = h \rho g$$

Where h = depth / height of fluid column, ρ = density of fluid, g = acceleration due to gravity [cite: 7].

If atmospheric pressure is P_i , the total pressure at depth h is $P_{\text{total}} = P_i + h \rho g$ [cite: 7].

Worked Example 6

A fish swims at a depth of 5 m in a reservoir (density of water = 1000 kg m^{-3} , $g = 10 \text{ m s}^{-2}$) [cite: 7].

- Pressure due to water: $P = h \rho g = 5 \times 1000 \times 10 = 50,000 \text{ Pa}$ [cite: 7].
- Total pressure (if atmospheric pressure = $1 \times 10^5 \text{ Pa}$): $50,000 + 100,000 = 150,000 \text{ Pa} = 150 \text{ kPa}$ [cite: 7].

5. Principle of U-Tube

Pressures at the same level in a continuous stationary liquid are equal. At the liquid-liquid interface in a U-tube [cite: 7]:

$$\text{Pressure on Left Arm} = \text{Pressure on Right Arm} \implies h_1 \rho_1 g = h_2 \rho_2 g$$

6. Pascal's Principle

When pressure is applied to an enclosed static uniform fluid, it is transmitted undiminished to every portion of the fluid and to the walls of the container [cite: 7]. Pressure remains constant throughout [cite: 7]:

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

Applications: Hydraulic jacks, Hydraulic lifts (Hoists) [cite: 7].

7. Archimedes' Principle

When a body is wholly or partially immersed in an incompressible, stationary, uniform fluid, the upward buoyant force (upthrust) acting on the body is equal to the weight of the fluid displaced by the body [cite: 7].

$$U = \text{Weight of Displaced Fluid} = m g = \rho V g$$

Where U = upthrust, V = submerged volume, ρ = density of fluid, g = gravitational acceleration [cite: 7].

8. Fluid Dynamics & Continuity Equation

Fluid flow is categorized into streamline (laminar) and turbulent flows [cite: 7]. In streamline flow, particle velocity at any point remains constant over time [cite: 7].

Equation of Continuity

The product of cross-sectional area and flow velocity remains constant along a streamline [cite: 7]:

$$A_1 V_1 = A_2 V_2 = \text{ext}\{Constant\}$$

9. Bernoulli's Principle

For an ideal, non-viscous, incompressible, uniform streamline fluid flow, the sum of pressure energy per unit volume, kinetic energy per unit volume, and potential energy per unit volume remains constant along any streamline [cite: 7]:

$$P + \frac{1}{2} \rho V^2 + \rho g h = \text{ext}\{Constant\}$$

Between any two points A and B on a streamline [cite: 7]:

$$P_A + \frac{1}{2} \rho V_A^2 + \rho g h_A = P_B + \frac{1}{2} \rho V_B^2 + \rho g h_B$$

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